NEUROPLANT INDUSTRIAL INTELLIGENCE LABS • PROCESS SAFETY DIVISION

Standard Operating Procedure & Process Safety Compliance Mandate

Document Ref: **SOP-TURB-088** • Revision: **4.1** • Effective Date: **December 2026** • Lifecycle Status: **ACTIVE MANDATORY**

Facility / Unit:	Unit 4 - Catalytic Cracking & High-Pressure Gas Compression	**Governing Standard:**	OSHA 29 CFR 1910.119 (PSM) & API 612 / API 617
Primary Asset:	Steam Turbine Drive ST-301A & Compressor C-301	**Secondary Codes:**	IEC 61511 (SIL-3 Loop) & ASME Sec VIII Div 1
Auxiliary Systems:	Lube Oil Console LO-301, Seal Gas Panel SGP-301	**Safety Classification:**	Critical Turbomachinery (Tier-1 Process Unit)
Key Valves:	Trip Valve TV-401A, Governor HCV-402, Anti-Surge FV-305	**Lead Specialist:**	Dr. Elena Rostova (Chief Turbomachinery Engineer)
Isolation Points:	Steam Casing V-501, Discharge Block V-502, Vent V-503	**Compliance Auditor:**	Marcus Vance (Safety Compliance Superintendent)

1. Scope, Thermodynamic Operating Context & Hazard Analysis

This Standard Operating Procedure governs mandatory operational safety, thermodynamic warmup, overspeed protection, and emergency trip isolation protocols for **Steam Turbine ST-301A** coupled to multi-stage **Centrifugal Compressor C-301**. The turbomachinery operates on superheated steam at **650 psig (44.8 barg) and 370°C (700°F)**, compressing volatile flammable hydrocarbon cracked gas (molecular weight 24.2, C1–C4 mixture) up to **380 psig** discharge pressure. Catastrophic process hazards include explosive gas release from dry gas seals, uncontained rotor overspeed burst (kinetic fragmentation), casing thermal distortion, and acoustic compressor surge resulting in catastrophic axial bearing failure.

2. Safety Instrumented System (SIS) & Interlock Specifications

In compliance with **IEC 61511 / ISA-84 (Safety Integrity Level 3 - SIL-3)**, the primary trip system utilizes 2-out-of-3 (2oo3) voting logic for turbine overspeed and compressor protection. The electronic overspeed trip system monitors magnetic pickups **SE-601A/B/C** and commands electro-hydraulic **Emergency Trip Valve TV-401A** to close within ≤ **300 milliseconds**. Dual-redundant **Suction Anti-Surge Recycle Valve FV-305** is modulated by dedicated high-speed controller UIC-305 to maintain a minimum safety margin of 12% above the aerodynamic surge limit line at all operating mass flow rates.

3. Pre-Start Permitting, Thermal Soak & Lubrication Prerequisites

Prior to initiating steam admission or turning gear disengagement, operating personnel must complete the following mandatory checks:

• **Lube Oil Circulation:** Auxiliary Lube Oil Pump **P-301B** must be energized, achieving minimum header pressure of **28 psig** on **PT-620** and bearing oil supply temperature stabilized between **40°C and 45°C**.

• **Dry Gas Seal Differential:** Seal gas supply pressure on **PDT-312** must maintain a positive differential of ≥ **50 psi** above compressor suction casing reference pressure with clean instrument Nitrogen.

• **Slow-Roll Thermal Soak:** Turbine rotor must be maintained on slow-roll rotation (800–1,000 RPM) for ≥ **45 minutes** via warming bypass valve **MOV-501B** with all casing steam traps verified draining freely. Rotor shaft bow eccentricity must indicate ≤ **0.8 mils pk-pk** on keyphasor probe **KP-601**.

SOP-TURB-088 • Emergency Shutdown, LOTO Protocol & Compliance Setpoints

4. Step-by-Step Emergency Shutdown & Isolation Sequence (LOTO Protocol)

Step 4.1 (Immediate Trip Actuation): Depress Emergency Stop Pushbutton **HS-301** in Unit 4 Control Room or locally at turbine front standard. Confirm **Trip Valve TV-401A** and Governor Valve **HCV-402** verify 100% closed status on DCS graphic within 1 second.

Step 4.2 (Anti-Surge Depressurization): Confirm anti-surge recycle valve **FV-305** drives to 100% full open failsafe position, equalizing compressor suction and discharge headers to prevent aerodynamic surge buffeting.

Step 4.3 (High-Pressure Steam Casing Isolation): Manually close high-pressure steam supply isolation gate valve **V-501**. Apply red Master Lockout Padlock #LOTO-ST-01 to handwheel chain hasp and secure tag.

Step 4.4 (Process Gas Header Isolation): Manually close compressor suction block valve **V-501A** and discharge block valve **V-502**. Lock each valve in fully closed position with approved Lockout Hasp #LOTO-C-02.

Step 4.5 (Depressurization & Flare Bleed): Open casing bleed valve **V-503** routed to high-pressure flare header. Confirm zero pressure (≤ 0.5 psig) on pressure transmitter **PT-625** and temperature reduction below 50°C before maintenance permit signoff.

Step 4.6 (Electrical Lockout): De-energize 4,160V main motor bus breaker **SWG-4B-07** and auxiliary lube oil pump motor **MCC-4C-02**. Execute live zero voltage verification at motor terminals using calibrated multimeter.

5. Quantitative Sensor Baselines, Alarm Limits & SIL-3 Trip Setpoints

Monitored Parameter	Sensor Tag	Normal Operating Range	High Alarm (Warning)	Emergency Trip (SIL-3)
Shaft Radial Vibration (X/Y)	VT-601A / VT-601B	0.8 – 1.8 mils pk-pk	2.5 mils pk-pk	**3.8 mils pk-pk (Trip)**
Thrust Bearing Metal Temp	TE-610A / TE-610B	65°C – 80°C (149–176°F)	95°C (203°F)	**110°C (230°F) (Trip)**
Lube Oil Supply Header	PT-620A / PT-620B	28 – 32 psig (1.9–2.2 barg)	22 psig (Low Alarm)	**16 psig (Low-Low Trip)**
Rotor Rotational Speed	SE-601A/B/C (2oo3)	7,800 – 8,400 RPM	8,820 RPM (105% MCS)	**9,240 RPM (110% Over-speed)**
Seal Gas Differential Pressure	PDT-312	60 – 85 psid	40 psid (Low Alarm)	**25 psid (Low-Low Trip)**

6. Active Procedural Contradiction Alert & Historical Incident Lessons

CRITICAL SAFETY CONTRADICTION & REGULATORY MANDATE:

Vendor Service Bulletin VSB-2026-11 permits rapid restarting of ST-301A within 20 minutes of hot trip if vibration is below 3.0 mils. **HOWEVER, PLANT SAFETY DIRECTIVE SOP-TURB-088 STRICTLY SUPERSEDES VENDOR ADVICE:** Hot restart prior to completing a mandatory **4-hour thermal cooldown and slow-roll rotation** is strictly prohibited under OSHA PSM 1910.119 regulations.
Historical Incident INC-2024-03: In March 2024, an unauthorized hot restart following an unverified trip resulted in permanent rotor shaft thermal bowing (4.2 mils), severe labyrinth seal destruction, and hydrocarbon leakage to the compressor deck. Bypassing the 4-hour thermal soak-down requirement will result in immediate revocation of operational certification.

7. Compliance Verification & Authorization Sign-Off

Role / Designation	Authorized Name & Title	Signature & Date	Status / Stamp
Lead Turbomachinery Engineer	Dr. Elena Rostova, PhD, PE	*E. Rostova* • 01-Dec-2026	**APPROVED**
Process Safety Superintendent	Marcus Vance, CSP	*M. Vance* • 01-Dec-2026	**VERIFIED SIL-3**
Unit 4 Operations Director	Arthur Pendelton	*A. Pendelton* • 02-Dec-2026	**ISSUED TO PLANT**